

Vegard Pettersen

Automation & Software Engineer

[Portfolio](#) | [GitHub](#) | vegardbp@gmail.com | [LinkedIn](#)

C++/C# · Python · TypeScript · DSP · Control Systems · GPU Compute



Selected Projects

[Portfolio](#) →

C25 Crane

Bachelor thesis · Seaonics · 2026

- Built a 1:25 four-axis crane replica with a team of four, using Seaonics' production control system. Developed the Python interface and most of the crane's mechanical and exterior design using CAD and Blender, and contributed to its 3D printing and mechanical assembly.

Dark Bros

Solo project · Unity · 2026

- Solo-developed 3D multiplayer game, including networking, gameplay systems, UI, shaders, 3D assets, Steam integration, and game design.

RC Car Simulator

NTNU group project · 2025

- Built a hardware-accurate simulator and control system for a Sphero RVR, including physics, 3D rendering, video streaming, and simulated sensors. Graded B

GPU Genetic Algorithm

NTNU exam project · 2025

- Built a GPU-accelerated genetic algorithm from scratch using HLSL compute shaders, with the fitness evaluation and selection pipeline running on the GPU. Graded A.

Additional projects: [GPU particle simulation](#) · [nonlinear fighter-jet control](#) · [C++ asteroid game](#)

[More project write-ups, interactive demos, and source links on the portfolio](#) →

Education

Bachelor of Engineering in Automation

2023–2026

NTNU, Ålesund

Average grade: **4.3 / 5.0**

A in Control Theory, Intelligent Systems, DSP, Mathematics, Physics, and OOP.

Experience

Staff

2025 · 6 months

Buldrehallen, Norway

Worked independently with responsibility for safety, security, customer service, POS transactions, and member administration. Gave safety briefings and technical instruction to new climbers.

Technical Skills

Languages: C++, C#, Python, C, GLSL, HLSL, WGSL, TypeScript

Frameworks & Technologies: Unity, React, WebGPU, OpenCV, Bullet Physics, Arduino

Tools: Git, Linux, vcpkg, Visual Studio, Blender, Fusion 360